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MILITARY SPECIFICATION

DETECTOR-DEWAR, DT-594/UA

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module, Detector-Dewar, DT-594/UA.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein. In the event of a conflict between the document referenced herein and the contents of this specification, the contents of this specification shall be considered a superceding requirement.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-C-49175	- Cooler, Cryogenic, Mechanical HD-1033B/UA and HD-1033C/UA
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-5855

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MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-726	- Packaging Requirement Codes
MIL-STD-756	- Reliability Modeling and Prediction
MIL-STD-810	- Environmental Test Methods

DRAWINGS

USACECOM

SM-C-806033	- Bellows, Cooler
54490-5004372	- Cryogenic Heat Load Assembly (1 Watt Integral)
SM-D-806561	- Detector-Dewar, DT-594/UA

2.2 Non-Government documents. The following document of the exact issue shown, forms a part of this specification to the extent specified herein. In the event of conflict between the document referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

Standard Procedures for Testing Infrared Detectors and Their Performance, R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Detector-Dewar is an opto-electronic device sensitive to radiation in the 7.5 to 12.0 micrometer spectral region. The detector-Dewar shall use a 180-element detector array of photoconductive mercury-cadmium-telluride operated from a supply of +5.0, ±0.2 volts direct current (Vdc). The detector bias resistors shall be an integral part of the detector-Dewar.

3.2 Construction. The detector-Dewar shall be in accordance with SM-D-806561 and as specified herein.

3.2.1 Weight. Weight of the detector-Dewar shall not be greater than 0.75 pound including bias resistors.

3.3 First article. When specified in the contract (see 6.2) the contractor shall furnish first article units for inspection (see 4.3).

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings.

3.5 Components. The detector-Dewar shall consist of a 180 element array and a resistor bias pack. The detector array orientation with respect to the Dewar mounting flange shall be variable through 360 degrees. At least two reusable, electrically activated getters shall be an integral part of the

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detector-Dewar for periodic maintenance of the vacuum, and each getter shall withstand at least seven firings after shipment. The getter activation current (amperes) and time (minutes) shall be provided on the label specified in SM-D-806561.

3.6 Performance characteristics.

3.6.1 Detector-Dewar defects. Unless otherwise specified herein, the detector-Dewar shall be operated at 80 (+5, -20) K. The detector-Dewar shall contain no more than 12 defective elements as defined in 6.3. Of the 12 defective elements, no more than 4 may be hard defects. In the central 18 elements of the array there shall be no hard defects and no more than 2 soft defects. The central 36 elements shall contain no hard defects and no more than 4 soft defects. A maximum of two hard defects may be located in elements 1-72 and a maximum of two may be located in elements 109-180. Hard defects shall be separated from one another by at least 4 elements and shall be electrically strapped to an adjacent element with common ground. Soft defects located within the central 36 elements shall be separated from any other defect by at least 4 non-defective channels. Soft defects shall not be strapped unless the strapping is from an adjacent hard defect at the end of a group of 5 commonly grounded elements. (See 6.3.2)

3.6.2 Time constant. The time constant of each detector element shall not be greater than 2 microseconds.

3.6.3 Sensitive area. The sensitive area of the detector elements shall fall within the dimensions specified on SM-D-806561.

3.6.4 Crosstalk. The crosstalk signal level between any two elements shall be less than -26 dB.

3.6.5 Specific Detectivity (D*). The 500 K blackbody specific detectivity of all nondefective elements shall be greater than 8.5×10^{9} cm $Hz^{\frac{1}{2}}W^{-1}$. The average specific detectivity of all elements of the array which are not hard defects shall be greater than 1.1×10^{10} cm $Hz^{\frac{1}{2}}W^{-1}$. (See 6.3.2)

3.6.5.1 Noise. The detector noise of each element shall be not less than 2.5×10^{-9} volts rms $Hz^{-\frac{1}{2}}$. (See 6.3.2)

3.6.6 Detector resistance. The detector resistance of each element shall be not less than 20 ohms and not greater than 150 ohms. (See 6.3.2)

3.6.7 Responsivity. The average of the 500 K blackbody responsivities of the acceptable elements shall be not less than 8.3×10^{3} volts/watt nor greater than 20×10^{3} volts/watt. The blackbody responsivity of each non-defective element shall not vary from the mean by more than +/- 20 percent of the mean value of the non-defective elements. (See 6.3.2)

3.6.8 Temperature sensors. At least two calibrated 2N2222 temperature sensor chips shall be installed at or near the detector array location in the Dewar to provide an electrical signal which is an analog of temperature. The temperature sensor shall have a junction voltage of 1.060, $\pm$0.015 volt with a 1 milliampere (mA) $\pm$.01mA forward current when the detector-Dewar is operated

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at 77, ±3 K. The voltage shall be recorded on the appropriate label on the detector.

3.6.9 Spectral response. Each element of the detector-Dewar shall have a spectral sensitivity of 7.7, ±0.3 to 11.7, ±0.5 micrometers.

3.6.10 Detector bias. The bias current shall not exceed 4.0 milliamperes on any single element. The total bias power dissipated by the array shall not exceed 100 milliwatts. The detector resistor bias pack shall be operated at +5.0, ±0.2 Vdc. Each detector element shall be biased with current flow in the same direction.

3.6.11 Dewar warm and cold shields. The clear aperture (warm shield) shall be in accordance with the detail requirements of SM-D-806561 and the following requirements: Cold shielding is optional (at the discretion of the detector manufacturer) but, when used, shall also conform to these same requirements.

a. In the direction of the array's axis (either (1) or (2))

(1) For each element, the cold shield(s) shall allow no vignetting of any ray to the center point on the detector surface within an angle of 18 degrees measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis.

(2) Each element shall be allowed to accept radiation from a 1.00, ±0.05 inch diameter stop centered on the detector-Dewar's optical axis and located a maximum of 1.60 inches behind the plane of the detectors.

b. In the direction perpendicular to the array's axis.

The cold shield(s) shall allow for each element, no vignetting of any ray to any point on the detector surface within an angle of 23 degrees measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis.

3.6.12 Heat load. The detector heat load shall not exceed 0.40 watt when all 180 detector elements are operating at their final bias current and the Dewar is cooled to 80, ±5 K.

3.6.13 Vacuum integrity. The Dewar shall comply with heat load requirements of 3.6.12 for a period of at least one (1) year.

3.6.14 Microphonics. When cooled to a temperature of 60-80 K, biased in accordance with paragraph 3.6.10, and vibrated in accordance with paragraph 4.6.14, all elements of the detector-Dewar shall have no noise peaks above the white noise reference value greater than 6db from 300 to 10,000 hertz, and 10db from 10,001 to 20,000 hertz. (See 6.3.2)

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3.6.15 Cooldown. When mated to a cryogenic cooler (MIL-C-49175) which is capable of cooling a test mass of 1440 (+25, -0) Joules of copper thermal mass (300 K to 80 K) (5004372) from ambient to 80 K in not less than 12 nor greater than 14 minutes, the dewar cooldown time, from ambient temperature to 80 K shall not exceed the time achieved with the test mass.

3.7 Environmental conditions.

3.7.1 Temperature shock. The detector-Dewar shall not be damaged (see 6.3.6) by sudden changes in temperature between -54°C and +71°C.

3.7.2 High temperature. The detector-Dewar shall not be damaged by storage or operation to +71°C.

3.7.3 Low temperature. The detector-Dewar shall not be damaged by operation to -54°C or storage to -62°C.

3.7.4 Shock. The detector-Dewar shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The detector-Dewar shall not be damaged by vibration over the frequency spectrum at the specified g levels and amplitudes shown in figure 1.

3.8 Reliability. The detector-Dewar shall have a specified mean time between failure (MTBF) of at least 27,600 hours.

3.9 Treatment and painting. Unless otherwise specified, the detector-Dewar shall be treated and painted in accordance with the applicable drawing.

3.10 Preconditioning. Each detector-Dewar shall be subjected to random vibration and temperature cycling in accordance with figure 2 prior to group A testing.

3.11 Nameplates and product marking. Unless otherwise specified, the detector-Dewar module, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection before packaging (see 4.5).
c. Packaging inspection (see 4.9).

4.3 First article inspection. When specified in the contract, the first article inspection shall be performed by the contractor on a minimum of 5 preconditioned (see 4.10) sample detector-Dewars.

4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and 4 percent AQL for visual inspection except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Marking not as specified	2.5	3.11	
Workmanship not as specified	2.5	3.12	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.9	
Dimensions not as specified	2.5	3.2	
Getters and label not as specified	1.0	3.5	SM-D-806561
Detector orientation range not as specified	1.0	3.5	SM-D-806561
Sensitive area not as specified	2/	3.6.3	SM-D-806561
Dewar warm and cold shields not as specified	2/	3.6.11	SM-D-806561

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.
2/ Inspect every array 100 percent and inspect at least 3 elements in each array.

4.3.2 Tests. First article inspection shall be conducted upon units which have successfully completed the inspections specified in 4.3.1. Tests may be conducted in any order, except that microphonics shall be performed after shock and vibration (4.7.4 and 4.7.5). Failure of any tests shall constitute first article failure.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Detector-Dewar defects	3.6.1	4.6.1
Time constant	3.6.2	4.6.2
Crosstalk	3.6.4	4.6.4
Detectivity	3.6.5	4.6.5
Detector resistance	3.6.6	4.6.6
Responsivity	3.6.7	4.6.7
Temperature sensors	3.6.8	4.6.8
Spectral response	3.6.9	4.6.9
Detector bias	3.6.10	4.6.10
Heat load	3.6.12	4.6.12
Vacuum integrity	3.6.13	4.6.13
Microphonics	3.6.14	4.6.14
Cooldown	3.6.15	4.6.15
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3
Shock	3.7.4	4.7.4
Vibration	3.7.5	4.7.5
Reliability	3.8	4.8

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.f).

4.4 Inspection procedures. Unless otherwise specified herein, the detector-Dewar shall meet the specified performance requirements at an ambient temperature of +23°C, ±5°C, when operated at a temperature of 80, ±5 K, and biased to +5.0, ±0.2 Vdc.

4.5 Quality conformance inspection before packaging.

4.5.1 Inspection. Inspection shall be specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted on all detector-Dewars following successful inspection of 4.5.1 and preconditioning of 4.10. Group A tests listed in Table III may be performed in any order except that detector-Dewar defects (4.6.1) shall be run last. Failure of any test shall be cause for rejection of that unit.

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TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Microphonics	3.6.14	4.6.14
Detectivity	3.6.5	4.6.5
Detector resistance	3.6.6	4.6.6
Responsivity	3.6.7	4.6.7
Temperature sensors	3.6.8	4.6.8
Detector bias	3.6.10	4.6.10
Heat load	3.6.12	4.6.12
Detector-Dewar defects	3.6.1	4.6.1

4.5.2.2 Group B inspection. Group B inspection is not required.

4.5.2.3 Group C inspection. Group C inspection shall be conducted on detector-Dewars selected from units which have passed the tests in 4.5.2.1. The tests of 4.6.9, 4.6.13 and 4.6.15 shall be performed on the sample quantity. The test of 4.7.5 shall be performed prior to the group C inspections on every third lot.

4.5.2.3.1 Sampling for group C inspection. The tests of 4.6.9, 4.6.13 and 4.6.15 shall be initially performed at a reduced sample rate of 3 percent of the specified lot (see 6.3.8) or one unit, whichever is the larger quantity.

a. When zero failure occurs, the 3 percent or one unit inspection sampling shall continue.

b. Failure of any unit to pass the inspection of either 4.6.9, 4.6.13 or 4.6.15 shall require 100 percent inspection of the entire lot for the failed attributes only with continued follow-on lots inspected until zero failure occurs in any lot. Upon achieving zero failure at 100 percent inspection, follow-on lot inspection shall be conducted at 10 percent sampling. Thereafter, upon achievement of zero failure during 10 percent sampling, the sample rate shall be lowered to 3 percent of the specified lot or one unit, whichever is larger. However, if one or more failures occur during the 10 percent inspection sampling, the 100 percent inspection testing shall be reinstituted.

4.5.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2).

4.5.2.3.3 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of Table III.

4.5.2.4 Group D inspection. Group D inspection not required.

4.6 Test methods.

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4.6.1 Detector-Dewar defects. The detector array shall be checked for defects. Failure to meet requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Time constant. Measurement shall be performed on 5 elements of the array each separated by at least 20 elements. The time constant is defined as the time required for the element response to reach 63 percent of its steady state value when irradiated by a 7.5 to 12 micrometer source which produces a signal-to-noise ratio not greater than 100:1. This time constant shall be considered equal to the time constant defined by the high frequency rolloff of the noise spectrum of the detector. Any measurement technique consistent with these definitions shall be considered acceptable. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Sensitive area. The physical area as defined by the attached drawing shall be determined by optical microscope examination. Measurement shall be made on 3 elements of the array each separated by at least 20 elements prior to the integration of the Dewar window with the detector-Dewar. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Crosstalk. Measurements shall be made on 4 elements of the array. The crosstalk shall be measured under the following conditions:

a. An element shall be irradiated by a 7.5 to 12 micrometer source having a signal level 20 times greater than the noise level for the element. The image of the source shall fall entirely within the area of that detector element.

b. The signal level of the element adjacent to and on the same ground path of the irradiated element shall be measured. The element under test shall meet the requirements of 3.6.4.

Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Blackbody specific detectivity (D*). Noise measurements shall be made on each element of the array under the following conditions: detector cooled to 77, $\pm$3 K with LN_2, 295, $\pm$5 K background temperature and equivalent noise bandwidth of 9 KHz $\pm$ 200 Hz with a low frequency 3dB point (high pass) no greater than 80Hz. Using the results of these measurements, and the nominal detector size (see 6.3.7), the 500 K blackbody specific detectivity for each element of the array shall be calculated. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Detector resistance. The resistance of each element of the array shall be measured. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Responsivity. Blackbody responsivity of each element of the array shall be measured when the detector is cooled to 77 $\pm$ 3 K. Nominal detector element size (see 6.3.7) shall be used in responsivity calculations and blackbody temperature shall be 500, $\pm$5 K. Failure to meet requirements of 3.6.7 shall constitute failure of this test.

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4.6.8 Temperature sensors. The 77 K calibration point shall be determined by cooling the inner Dewar with liquid nitrogen. One milliampere ± .01 milliampere bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77 K calibration point. Failure to meet requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Spectral response. The spectral response of element numbers 6, 91, and 176 shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R. C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these are hard defects, and adjacent element may be substituted. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled detector-Dewar. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Detector bias. The detector bias power shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R. C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. Failure to meet requirements of 3.6.10 shall constitute failure of this test.

4.6.11 Dewar warm and cold shields. During in-process inspection, the warm and cold shields shall be measured to ensure conformance with the requirement. Alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection. Failure to meet requirements of 3.6.11 shall constitute failure of this test.

4.6.12 Heat load. The heat load of the detector-Dewar shall be determined with all detectors operating at specified bias. The heat load of the detector-Dewar shall be determined by partially filling the Dewar's inner stem with liquid nitrogen and monitoring the rate of loss of the cryogen. The heat load shall be calculated from the average or integrated loss rate of the cryogen for the last 5 minutes of the Dewar's hold-time. For calculations, cooling produced by the nitrogen phase change only shall be considered as total useful energy. If this test is performed with the detector array unbiased, the results of this test shall be added to the results of the detector bias power test to determine the heat load. Failure to meet requirements of 3.6.12 shall constitute failure of this test.

4.6.13 Vacuum integrity. Heat load shall be measured on the unbiased detector-Dewar. The temperature of the detector-Dewar shall then be elevated to +71°C for a period of 72 hours. At the completion of the temperature exposure and stabilization at ambient temperature, the heat load shall be measured again. Getters may not be fired during this test. Failure of the detector-Dewar to meet the requirements of 3.6.12 or a 15 percent increase in heat load between the measurement before and after temperature exposure shall constitute failure of this test.

4.6.14 Microphonics. Microphonics shall be measured while the detector is vibrated in accordance with paragraph 4.6.14.1. The detector-Dewar shall be cooled to 60-80 K with full operating bias applied. Mounting to the cooler shall be made by tightening the mounting screws to a torque of 10.00 ± 2.0 inch pounds. Vibrational excitation shall be measured by an accelerometer

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mounted on the mounting plate of the detector-Dewar. The white noise reference level is determined by averaging the noise on each channel from 12000 to 20000 hertz with vibration applied. Noise spectra shall be measured using a bandwidth between 50 hertz and 365 hertz. Failure of any channel to meet the requirements of 3.6.14 shall constitute failure of this test.

4.6.14.1 Microphonics Vibration. The detector-Dewar shall be coupled to a mechanical cryogenic refrigerator (HD-1033) that produces random vibrational excitation of the detector/Dewar in three mutually perpendicular axis and cools the dewar to 60-80 K. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 hertz. The RMS sum of the three axis vibration shall be at least 2.0 G RMS. Each axis shall produce a minimum vibrational excitation of 0.7 G RMS. In the frequency range of 2,000 to 20,000 hertz, the RMS sum shall be at least 1.7 G RMS. The cooler shall be mated to the detector with a metal bellows (P/N SM-C-806033).

4.6.15 Cooldown. An HD-1033 cryogenic cooler shall be mated to the standard test mass (5004372). The capacity of the HD-1033 shall be such that the test mass is cooled from ambient (296 K) to 80 $\pm$ 1½ K in not less than 12 nor greater than 14 minutes. This measurement shall be made using highly polished metal Dewar without any internal wrapping, a vacuum of less than 1 x 10^{-5} torr and constantan leadwires which are a maximum of six inches in length and a resistance less than one percent of the value of the resistor in the cold station test mass. The time to cool the thermal mass shall be recorded. The voltage output of the temperature sensor diodes in the detector-Dewar under test shall be measured at 77 $\pm$ 1 K and at ambient (296 K). Using these two points the voltage output at 80 K shall be interpolated. The cooler shall then be coupled to the detector using thermal grease and the Bellows, Cooler (SM-C-806033). Operating bias power shall be applied and the time for the temperature of the diodes to reach 80 K shall be measured and recorded using the previously plotted diode calibration curve. Failure of the diodes to cool to within the requirements of 3.6.15 shall constitute failure of this test.

4.7 Environmental tests. The environmental tests shall be performed in accordance with MIL-STD-810 as modified by this specification. The performance tests required by MIL-STD-810 before and after each environmental test shall be those listed in table III. During environmental test, the units need not be energized or cooled down but shall be subjected to a continuity check for each element.

4.7.1 Temperature shock. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 503.1 Procedure I except that the temperature of step 2 shall be -54°C. Failure to meet requirements of 3.7.1 and Table III shall constitute failure of this test.

4.7.2 High temperature. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 501.1 Procedure I. The Temperature of steps 4 and 5 shall be +71°C. Failure to meet requirements of 3.7.2 and Table III shall constitute failure of this test.

4.7.3 Low temperature. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 502.1 Procedure I except that the storage temperature of step 2 shall be -62°C and the low operating temperature of step 4 shall be

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-54°C. Failure to meet requirements of 3.7.3 and Table III shall constitute failure of this test.

4.7.4 Shock. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-1 (100 g's peak value at 11 milliseconds time duration) and Procedure V. Continuity checking during shock testing is not required. Failure to meet requirements of 3.7.4 and Table III shall constitute failure of this test.

4.7.5 Vibration. The detector-Dewar shall be tested in accordance with MIL-STD-810, Method 514.2, Procedures I and VIII except that the curve shown in figure 1 of this specification shall be used in lieu of the MIL-STD-810 curves. The sinusoidal cycle time shall be 120 minutes/axis. The dwell time shall be 1/6 of the cycling time at each axis. Failure to meet requirements of 3.7.5 and Table III shall constitute failure of this test.

4.8 Reliability. A reliability prediction and math model shall be maintained by the contractor in accordance with MIL-STD-756B task sections 100 and 200 which reflect the current design and components of the detector-Dewar. Failure of the prediction or model to explicitly delineate compliance with paragraph 3.8 shall constitute failure of the reliability requirement.

4.9 Inspection of packaging. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.9.1 First article inspection of packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.9.2 Quality conformance inspection of packaging.

4.9.2.1 Materials inspections. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.9.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.9.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table IV, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

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TABLE IV. PACKING INSPECTION PROVISIONS.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight and Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.10 Preconditioning. Each detector-Dewar shall be subjected to preconditioning environments 3.10 and figure 2.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-1-00-00-NU-X-XX-P-ED-A.

5.1.2 Level B. 2M-1-1-00-00-NU-X-XX-P-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

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6.1 Intended use. The detector-Dewar in intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required.

(1) Time frame for submission of first article test reports when first article tests are required.

(2) Time frame for approval of first article test reports when first article tests are required.

(3) First article reliability testing if required.

c. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to group C failures (see 4.5.2.3.2).

d. Level A or level B preservation and packaging (see section 5).

e. MIL-STD-810C shall be used for environmental tests of 4.7.

f. Deliver all first article samples to - Commander, USACECOM, ATTN: AMSEL-NV-SE, Fort Belvoir, VA 22060-5677.

g. The microphonic requirement, 3.6.14, is required for procurement of all modules to this specification made directly by the Government. The requirement, 3.6.14, may be waived by system contractors ordering modules to this specification for use in the manufacture of completed systems if these systems meet system performance requirements which include an equivalent microphonics check.

h. MIL-STD-756B shall be used for the reliability prediction (see 4.8).

i. Detector-Dewar drawing number(s).

j. When first article rough handling tests are not required.

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.2.1 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.2.2 Environmental. Environmental pollution measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 Definitions.

6.3.1 Element. An element is defined as one channel in the 180 element detector-Dewar.

6.3.2 Defective element. A defective element shall be defined as an element which fails to meet any of the requirements of 3.6.5, 3.6.5.1, 3.6.6, 3.6.7 and 3.6.14. A defective element may be classified as either hard or soft. A hard defect is an element which fails the requirements 3.6.5.1, 3.6.6, 3.6.7, or 3.6.14 or which has a 500 K blackbody specific detectivity less than 6.0×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$. Two elements shorted together are considered one hard defect. A soft defect is one which meets all requirements of 3.6.6, 3.6.7, and 3.6.14 and has a 500 K blackbody specific detectivity equal to or greater than 6.0×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$, but less than 8.5×10^9 cm $Hz^{\frac{1}{2}}W^{-1}$.

6.3.3 Strapping. Strapping is defined as electrically connecting a defective element to a nondefective element. The pair of strapped elements thus formed, exhibiting 100 percent crosstalk, shall be considered as a single defect.

6.3.4 Array axis. The array axis is defined as a line on the surface of the array passing through the center of the projected geometrical areas of each element.

6.3.5 Optical axis. The detector-Dewar's optical axis is defined as a line perpendicular to the detector mounting plate that intersects the array axis at the mid-point of the array.

NOTE: One-half of the active detector elements are located on each side of the optical axis.

6.3.6 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of detector-Dewar performance to values less than specified herein.

6.3.7 Nominal detector element size. Defined as 0.0016 by 0.0023 inch.

6.3.8 Specified lot quantity. The specified lot quantity shall be one-month's production unless the contract specifies a different lot basis such as one-week's production or those units processed as a single batch.

6.3.9 Detector Temperature. Detector temperature is defined as that indicated by the temperature sensor diode.

6.3.10 Existing Suppliers. Suppliers who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12-month period. If a supplier cannot maintain monthly average, moves their production facilities, has significant management/personal changes, and/or has significant changes in production methods-

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tooling-processes, a first article test (less reliability) performed annually shall sustain the existing supplier's status when performed on five (5) units manufactured by the supplier explicitly for this test. Upon implementation of this specification, suppliers who are then supplying equipment to specification B2-28A050102, MIL-D-49172, or MIL-D-49457 with the minimum monthly delivery rate specified above, shall be considered as existing suppliers.

6.4 Variation of responsivity with temperature. When coupled to a cryogenic refrigerator, the temperature of the array may reach 55 K. Under those conditions the responsivity (paragraph 3.6.7) may decrease to as low as 5.6 x 10^3 volts/watt.

CUSTODIAN
Army - CR

PREPARING ACTIVITY
Army - CR

Project 5855-A276

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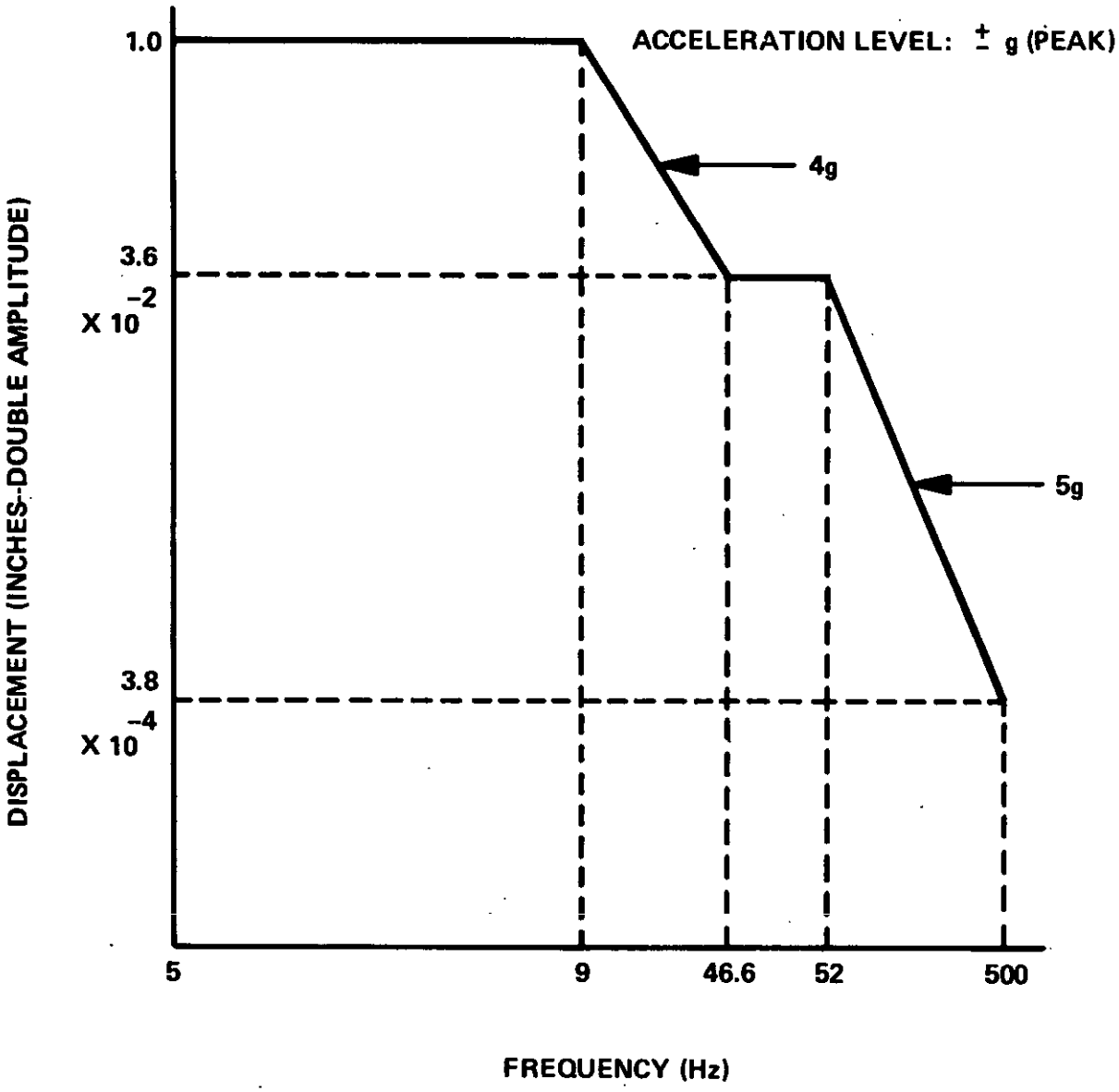
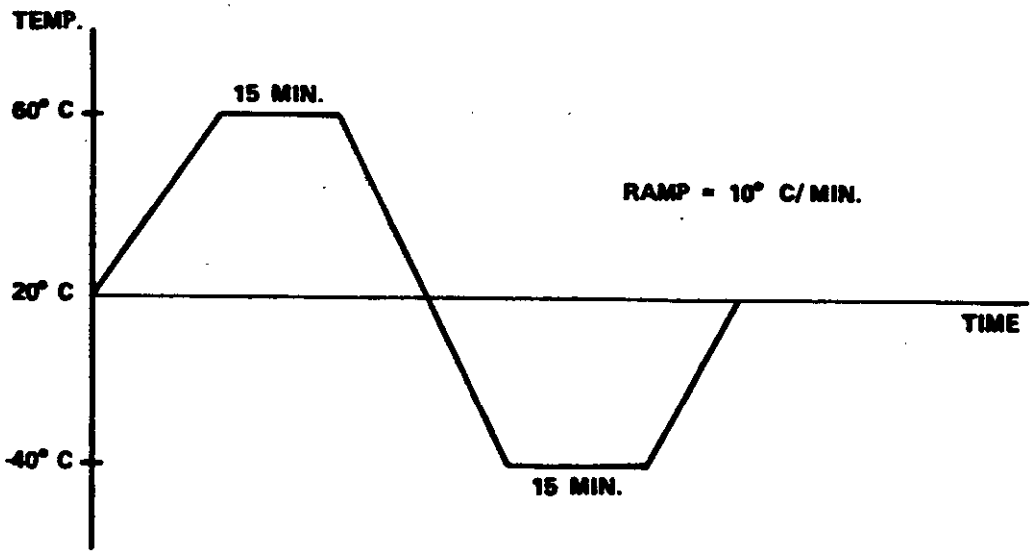


FIGURE 1. VIBRATION TEST PROFILE

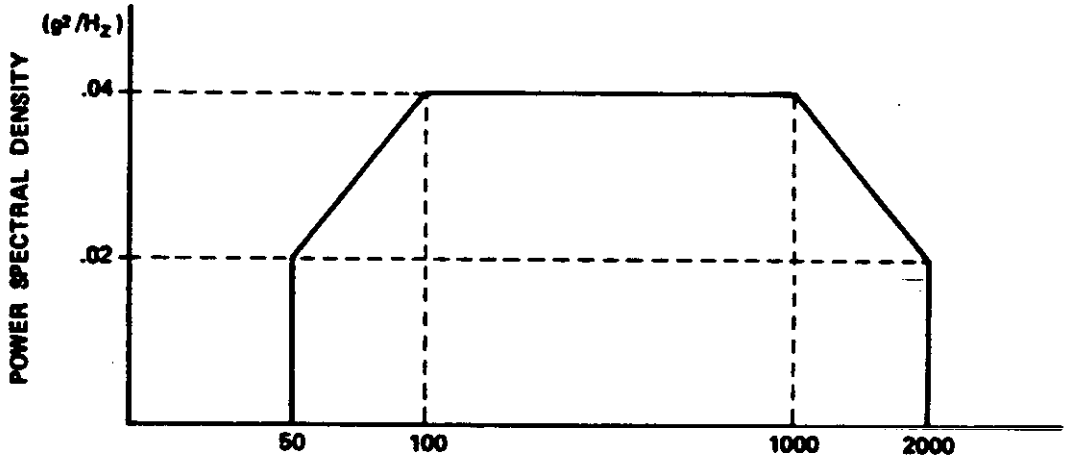
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TEMP CYCLE



VIBRATION CYCLE[1]

15 MIN



NOTES:
1. VIBRATION PERFORMED ALONG DETECTOR COLD WELL AXIS.
2. TEMP. CYCLE PRECEEDS VIBRATION CYCLE.

FIGURE 2. Preconditioning cycle

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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